

# Hari Prasath S

SOFTWARE DEVELOPER · AI ENGINEER  
INFERENCE ENGINEERING · DISTRIBUTED SYSTEMS

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## EXPERIENCE

### Agent Platform

- **Co-developed an LLMOps platform layer** on Agno and LiteLLM, separating agent orchestration from provider access and centralizing LLM APIs, LLM observability and security in one gateway.
- **Built a stateless tool server with FastMCP and FastAPI**, enabling agent tool calling for user-authored Python tools by dynamically provisioning short-lived MCP servers — horizontally scalable without per-tool Lambdas.
- **Architected and built an AI governance layer for PII protection**, de-identifying personal data before LLM calls and restoring it for tools and responses, preserving conversational coherence.
- **Built real-time voice agents on LiveKit Agents** that join and converse in live meetings via STT, TTS, and voice activity detection, with speech usage metering.
- **Built an agentic NL2SQL analytics assistant with Google ADK and Gemini**, translating natural-language questions into parameterized SQL and synthesizing results into actionable insights.

### Retrieval & Evaluation

- **Built multimodal RAG for AI agents** using embeddings and Qdrant to retrieve relevant context from user-provided data, grounding agent responses and tool use in that context.
- **Built LLM-as-a-Judge model evaluation for AI agents**, benchmarking candidate models on accuracy, bias, and fairness to guide model selection.

### Inference Engineering

- **Re-engineered the inference path** around a persistent vLLM engine and streaming relay, replacing per-request AWS Step Functions/Batch execution — reducing time-to-first-token from **1.9 s to 23 ms (~99%)**, eliminating gateway timeouts, and accelerating repeat agent calls **24x** through cached prompt prefixes.
- **Built autoscaling multi-model serving on ECS**, packing multiple vLLM engines per GPU to cut per-model cost by **75%**; drove scale-out/in via custom CloudWatch metrics and alarms, with health-gated production deployments.

### Engineering Practice

- **Resolved production issues end-to-end** — from root-cause analysis through fixes and regression tests, with runbooks enabling reliable handoff and ongoing maintenance.
- **Drove system design and code reviews across these systems**, working cross-functionally with ML, product and customer teams; mentored an intern and made AI-assisted development an integral part of the engineering workflow.

## PERSONAL PROJECT · DELTA



- **Built Delta, a change-intelligence platform** that maintains live state per tracked entity and emits events only for material changes.
- **Engineered Delta's event-driven backbone with transactional outbox and Kafka**; added Debezium CDC for analytics/reporting and deployed services on Kubernetes with Helm, Terraform, and CI.
- **Kept Delta's state deterministic** — constrained LLM extraction to structured output grounded in source text at ingest, while RAG chat cites retrieved evidence at query time.

## TECHNICAL SKILLS

|                  |                                                                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LANGUAGES        | Python · Java · TypeScript · JavaScript · SQL · Bash                                                                                                               |
| GENAI & LLMOPS   | Agno · LiteLLM · Model Context Protocol · FastMCP · Function Calling · RAG · Prompt Engineering · Context Engineering · Model Evaluation · Amazon Bedrock · Gemini |
| AI GOVERNANCE    | Guardrails · PII Protection · Data Privacy · Hallucination Mitigation · AgentOps                                                                                   |
| ML & MLOPS       | vLLM · PyTorch · CUDA · KV Cache · Model Deployment · Model Monitoring · GPU Benchmarking                                                                          |
| RETRIEVAL & DATA | PostgreSQL · pgvector · Vector Databases · Reranking · Embeddings · Redis · MongoDB · OpenSearch                                                                   |
| BACKEND & APIS   | FastAPI · Pydantic · asyncio · SQLAlchemy · Celery · Kafka · REST · SSE · Microservices                                                                            |
| CLOUD & DEVOPS   | AWS (ECS · EC2 · Lambda · Step Functions · SQS · S3 · CloudWatch) · Kubernetes · Helm · Terraform · Docker · CI/CD · Linux                                         |